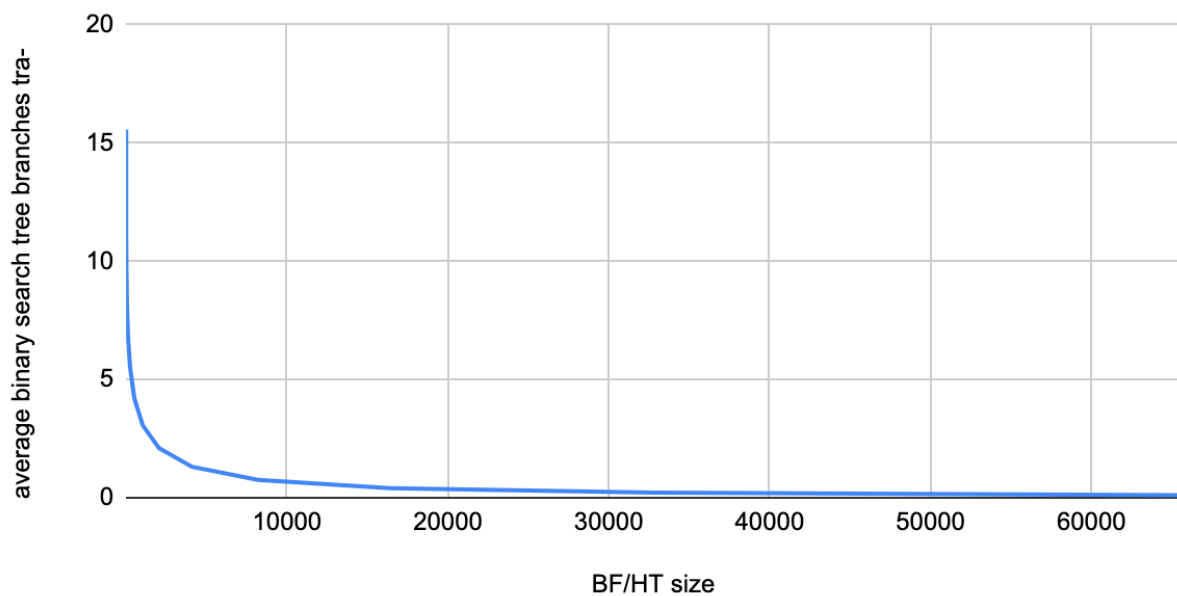


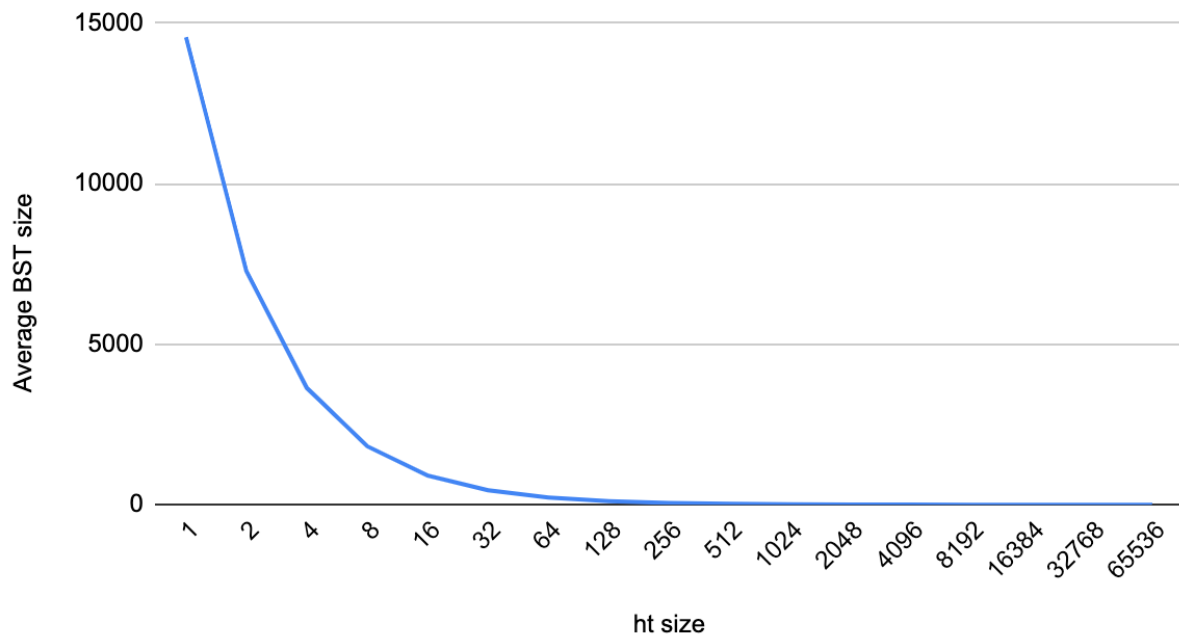
Assignment 7 Writeup: The Great Firewall of Santa Cruz - Bloom Filters, Linked Lists, Binary Trees, and Hash Tables

This graph compares the average binary search tree branches traversed as the bloom filter and hashtable size increases at the same rate.

average binary search tree branches traversed vs. BF/HT size

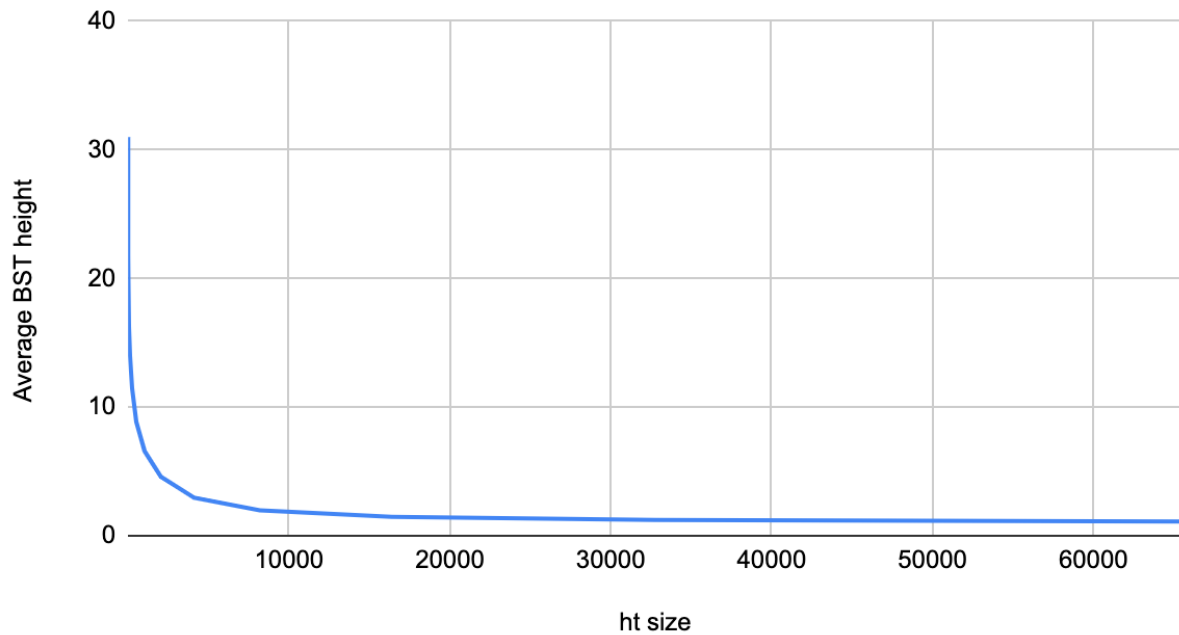


Average BST size vs. ht size



This graph shows that as the bloom filter and hash table size increases, the number of lookups increases exponentially. The average binary search tree branches traversed is calculated by dividing the number of branches by the number of lookups. Thus, as the size decreases, the average binary search tree branches traversed decreases exponentially.

Average BST height vs. ht size



The heights of the binary search trees stay the same if you adjust the bloom filter size, however they change when you alter the hash table size because of collisions. As you increase the hash table size, the average binary search tree height decreases exponentially.